Test category	Type Test		
Type	LNA426_16SP_WE, KBN426_16S_WE_G3	**Test Method**	SNI IEC 60884-1: 2014
Ranking	16A 250V~	**Job Order No.**	WO/25110097
Test Date	25/11/2025	**Request No.**	R/25110097
Room Temp.	24	**Sample code**	A B C
Humidity	53	**Test Equipment No.**	QE/14/PH/33, QE/14/PH/36, QE/14/PH/53, QE/14/PH/55, QE/14/EL/26

Clause	Test requirements	Check up result	Decision
12	**TERMINAL AND TERMINATION**		**PASSED**
12.1	**General**		
	All tests on the terminal, with the exception of the tests in subclause 12.3.11, are carried out after the tests in clause 16.	All tests on the terminal, withthe exception of sub-testsclause 12.3.11, is carried out aftertesting of clause 16	L
12.1.1	Fixed rewiring socket outlets shall be equipped with screw type terminals or with screwless terminals.	Fixed contact box equipped with screw type terminals	L
	Portable rewiring plugs and socket outlets shall be equipped with terminals with screw clamps	-	TB
	If pre-soldered flexible conductors are used, care must be taken that in screw-type terminals the presoldered area must be outside the clamping area when connected as per normal use.	-	TB
	The means of clamping the conductors in the terminal shall not serve to fix other components, although such clamps may hold the terminal in place or prevent it from rotating.	The conductor clamping means do not servethe purpose of fixing other components.	L
12.1.2	Non-rewireable accessories must be equipped with solder, weld, clamp (crimp) or a permanent relationship of equivalent effectiveness (termination);	-	TB
	Screw or stud connections shall not be used.	-	TB
	The connection is made by clamping (crimp) previously soldered flexible conductors are not permitted, unless the solder area is outside the clamp area (crimp)	-	TB
12.2	**Terminals with screw clamps for external copper conductors**		
12.2.1	**The accessories must be equipped with terminals that allow perfect connection of the copper conductors:**		
	Rated current (A); Type of equipment;	16A ; Magun	
	Conductor type (rigid/flexible);	Rigid	
	Smallest/largest cross-sectional area (mm)²	1.5 and 2.5mm²	
	Largest conductor diameter (mm);	2.13mm²	
12.2.2	Terminals with screw clamps shall allow the conductors to be connected without special preparation.	Fulfil	
12.2.3	Terminals with screw clamps must have adequate mechanical strength.	Fulfil	
	Screws and nuts for clamping conductors have	Fulfil	

	ISO metric threads or threads of comparable depth (pitch) and its mechanical strength		
	Screws should not be made of soft or easily broken metal (creep), such as zinc and aluminum	Fulfil	
12.2.4	**Screw clamp terminals must be resistant to corrosion**		
12.2.5	Terminals with screw clamps shall be designed and constructed so that they can clamp the conductor without causing undue damage to the conductor	The terminals show no damage when the conductor is clamped.	
	Torsion test (15 minutes):		
	- conductor type;	Rigid	
	- number of conductors;	1	
	smallest cross-sectional area (mm²); bushing hole diameter (mm); height H (mm); mass (kg)	1.5; 6.5; 260; 0.4	
	- largest cross-sectional area (mm²); bushing hole diameter (mm); height H (mm); mass (kg)	2.5; 9.5; 280; 0.7	
	- nominal thread diameter (mm); screwdriver torque (Nm)	3.86mm; 1.2Nm	
	During the test, the conductor must not become detached from the clamp unit or break near the clamp unit, nor shall the conductor become damaged to such an extent as to render it unfit for further use.	Fulfil	
12.2.6	Terminals with screw clamps must be able to clamp conductors reliably between metal surfaces.	Terminals are reliably connected between metal surfaces	
	Tensile test (1 minute):		
	- conductor type;	Rigid	
	- number of conductors;	1	
	- smallest cross-sectional area (mm²); drawn (N)	1.5; 40	
	- largest cross-sectional area (mm²); drawn (N)	2.5; 50	
	- nominal thread diameter (mm); torque 2/3 x 1.2 (Nm)	0.8Nm	
	During the test, the conductor must not appear to move within the terminal	Fulfil	
12.2.7	Terminals with screw clamps shall be designed or located so that none of the rigid solid conductors or stranded conductor wires can be dislodged when the screws and nuts are tightened.	The conductor does not come out when screws tightened	
	- fixed contact box:	Solid rigid; 2.5mm²	

	rigid solid copper conductor/rigid stranded conductor with the largest cross-sectional area (mm^2);		
	- plugs and portable sockets: flexible conductors with the largest cross-sectional area (mm^2);	-	
	- terminals intended for winding two or three checked conductors, installed with the permitted number of conductors;	-	
	- number of wires (n) and nominal diameter of conductor (n x mm);	1 x 1.78	
	nominal thread diameter (mm); torque 2/3 x 1.2 (Nm)	0.8Nm	
	After testing, there must be no conductor wire that comes loose outside the clamp unit	Fulfil	
12.2.8	Terminals with screw clamps shall be fixed or placed in the fittings so that, when the clamping screw and nut are tightened or loosened, the terminal will not become loose from the fittings in the fittings.	Terminal is not loose when installation	
	Torque test:		
	- rigid solid copper conductor with the largest cross-sectional area (mm^2);	$2.5mm^2$	
	- nominal thread diameter (mm); torque (Nm)	3.86mm; 1.2Nm	
	- screws and nuts are tightened and loosened 5 times. During the test, the terminal must not be loose and there must be no damage such as broken screws or damaged screw heads, threads, washers that will be detrimental to the further use of the terminal.	The terminal is not loose and does nothaving damaged	
12.2.9	The clamping screw and nut of the earth terminal with the screw clamp must be locked sufficiently against possible loosening and must not be possible to loosen without the aid of tools.	Fulfil	
12.2.10	The earthing terminal with screw clamp must be such that there is no risk of corrosion.	Fulfil	
	The earth terminal body must be brass or other metal that is not less resistant to rust	Fulfil	
	If the earthing terminal body is a frame or enclosure part of an aluminum alloy,	-	

	precautions must be taken to avoid the risk of rust corrosion.		
12.2.11	For pillar terminals, the distance between the clamping screw and the end of the conductor, when fully inserted, shall be at least equal to that specified in figure 2: required (mm); measured (mm):	-	
	For mantle terminals: the distance between the fixed part and the end of the conductor, when fully inserted, shall be at least as specified in figure 5: required (mm); measured (mm):	-	
12.3	**Screwless terminals for external copper conductors**		
12.3.1	**Screwless terminals may be the suitable type for:**		
	- rigid copper conductor or	-	
	- both rigid and flexible copper conductors (tests were performed with rigid and then repeated with flexible conductors)	-	
12.3.2	Screwless terminals shall be equipped with two clamping units that allow suitable connection to rigid conductors or both rigid and flexible copper conductors having a nominal cross-sectional area	-	
	Rated current (A);	-	
	Conductor type (rigid / flexible) ;	-	
	Nominal cross-sectional area (mm²) ;	-	
	Largest diameter of rigid conductor (mm);	-	
	Largest diameter of flexible conductor (mm);	-	
12.3.3	Screwless terminals must allow conductors to be connected without special preparation	-	
12.3.4	The screwless terminal section is intended to carry current with the material as specified in 26.5	-	
12.3.5	Screwless terminals shall be designed so that they clamp the specified conductor with sufficient contact pressure without causing damage to the conductor. The conductor shall be clamped between metal surfaces.	-	
12.3.6	It must be clear how the connection and disconnection of	-	

	conductors is carried out.		
	Disconnection of conductors shall require an action, other than pulling the conductor, such that it can be performed manually with or without the aid of tools for general use.	-	
	There shall be no possibility of confusion regarding the holes for the use of tools to assist connection or disconnection, with the holes for the conductors.	-	
12.3.7	**Screwless terminals are intended for the interconnection of two or more conductors and are designed to:**		
	- during installation, the operation of the clamping means of one conductor is independent of the operation of the other conductor;	-	
	during disconnection, the conductors can be disconnected either simultaneously or separately;	-	
	- each conductor must be inserted in a separate clamping unit (no need for a separate hole);	-	
	- it must be possible to safely clamp a number of conductors up to the maximum number according to the design.	-	
12.3.8	Fixed socket-outlet screwless terminals shall be so designed that adequate conductor insertion is evident and excessive conductor insertion is prevented.	-	
12.3.9	Screwless terminals must be properly fixed to the socket.	-	
	Screwless terminals must not be loose when conductors are connected or disconnected during installation	-	
12.3.10	Screwless terminals must withstand the mechanical stresses occurring in normal use	-	
	Conductors connected and disconnected 5 times: 2.5 mm rigid solid copper conductor2	-	
	Conductors connected and disconnected 5 times: 1.5 mm rigid solid copper conductor2	-	
	After each connection, the conductor was subjected to a pull of 30 N for 1 minute with the pull applied without jerking.	-	

	During pulling, the conductor must not protrude from the screwless terminal.		
	Conductors connected and disconnected once: 2.5 mm rigid stranded copper conductor2	-	
	Conductors connected and disconnected once: 1.5 mm rigid stranded copper conductor2	-	
	After each connection, the conductor was subjected to a pull of 30 N for 1 minute with the pull applied without jerking. During pulling, the conductor must not protrude from the screwless terminal.	-	
	Screwless terminals intended for rigid and flexible conductors:		
	Conductors connected and disconnected 5 times: 2.5 mm flexible conductor2	-	
	Conductors connected and disconnected 5 times: 1.5 mm flexible conductor2	-	
	After each connection, the conductor was subjected to a pull of 30 N for 1 minute with the pull applied without jerking. During pulling, the conductor must not protrude from the screwless terminal.	-	
	For fixed socket outlets with screwless terminals, each conductor is subjected to a torsion test (15 minutes):		
	- conductor type;	-	
	- number of conductors;	-	
	- 1.5 mm^2; bushing hole diameter 6.5 mm; height H 260 mm; mass 0.4 kg	-	
	- 2.5 mm^2; bushing hole diameter 9.5 mm; height H 280 mm; mass 0.7 kg	-	
	During testing the conductor shall not appear to move within the clamping unit	-	
	After this test, no terminals or clamping means should be loose and the conductor should show no deterioration that would impair future use.	-	
12.3.11	Screwless terminals must withstand the electrical and thermal stresses that occur in normal use.	-	
	a) The test is carried out by loading the screwless terminal for 1 hour with alternating current and connecting a rigid solid conductor 1 m long.		
	Rated current (A);	-	
	Test current (A);	-	

	Conductor cross-sectional area (mm)2)	-	
	Number of screwless terminals	**1** **2** **3** **4** **5**	
	Voltage drop should not exceed 15 mV	10 20 30 40 50	L
	b) Screwless terminals that have been subjected to the voltage drop determination specified in test a) are previously tested as follows:		
	Rated current (A);	-	TB
	Test current (A);	-	TB
	Conductor cross-sectional area (mm)2)	-	TB
	The terminal is subjected to 192 temperature cycles, each temperature cycle has a period of approximately 1 hour with current flowing for approximately 30 minutes and then for approximately 30 minutes no current flows	-	TB
	Number of screwless terminals	**1** **2** **3** **4** **5**	
	Voltage drop is measured after 24 cycles.th	11 22 33 44 55	L
	Voltage drop measured after 48 cyclesth	- - - - -	TB
	Voltage drop measured after 72 cyclesth	66 77 88 99 00	L
	Voltage drop measured after 92 cyclesth	01 02 03 04 05	L
	Voltage drop measured after 120 cyclesth	06 07 08 09 10	L
	Voltage drop measured after 144 cyclesth	11 12 13 14 15	L
	Voltage drop measured after 168 cyclesth	16 17 18 19 20	L
	Voltage drop measured after 192 cyclesth	- - - - -	TB
	Voltage drop should not exceed 22.5 mV or twice the value measured after the 24th cycle, whichever is smaller	-	TB
	After this test, examination with normal vision or using glasses without the addition of a magnifying glass shall show no obvious changes that would interfere with further use such as cracks, deformation or the like.	-	TB
	In addition the mechanical strength test according to 12.3.10 is repeated and all specimens must withstand this test.	-	TB
12.3.12	**Screwless terminals shall be designed so that the rigid solid conductor to which they are connected remains clamped, even if it has been bent during normal installation.**		**PASSED**
	Compliance is checked by the following tests carried out on three switch specimens that have not been used for other tests.		
	Bending test:		
	Rated current;(A)	-	TB
	Style; (N)	-	TB
	Conductor cross-sectional area; (mm2)	-	TB

	Required voltage drop; (mV)	-			TB
	Number of complete specimens;	**1**	**2**	**3**	
	Starting point; (X = starting point of bending)	X	X+10°	X+20°	L
	Measurement of bending voltage drop to 1st	444	555	666	L
	Measurement of the 2nd bending voltage dropand	777	888	999	L
	Measurement of the 3rd bending voltage dropth	101	102	103	L
	Measurement of the 4th bending voltage dropth	104	105	106	L
	Measurement of the 5th bending voltage dropth	107	108	109	L
	Measurement of the 6th bending voltage dropth	1	2	3	L
	Measurement of the 7th bending voltage dropth	4	5	6	L
	Measurement of the bending voltage drop to 8th	7	8	9	L
	Measurement of the 9th bending voltage dropth	10	11	12	L
	Measurement of bending voltage drop to 10th	13	14	15	L
	Measurement of bending voltage drop to 11th	16	17	18	L
	Measurement of the bending voltage drop to 12th	19	20	21	L
	Voltage drop should not exceed 22.5 mV or twice the value measured after the 24th cycle, whichever is smaller	-			TB
	After this test, examination with normal vision or using glasses without the addition of a magnifying glass shall show no obvious changes that would interfere with further use such as cracks, deformation or the like.	-			
	In addition the mechanical strength test according to 12.3.10 is repeated and all specimens must withstand this test.	-			

Test case decision description:

- Testing meets requirements : Pass (L)
- Testing does not meet requirements : Not Passed (TL)
- Testing does not apply to test objects : Not applicable (TB)

PT Qualis Indonesia	Witnessed by		
	Name	Company	Signature
Tested by:			
Checked by:			